

AC-50 Air Compressor

Operating Limits and Diagnostic Reference

SIMULATED TRAINING DOCUMENT - NOT A MANUFACTURER MANUAL

1. Equipment Identification

Equipment code	AC-50
Equipment type	Oil-lubricated rotary compressor
Service	Workshop compressed-air training system
Normal discharge pressure	6.5-8.0 bar
High-temperature warning	Above 100 C
Receiver limit	10 bar maximum
Manual revision	Training Revision 1.0

System Overview

The AC-50 contains an electric motor, compression element, inlet filter, separator, cooler, receiver, drain and control valves.

SAFETY PREREQUISITES

Shut down the compressor. Isolate electrical power and apply lockout/tagout. Close the outlet valve. Vent receiver pressure to zero and verify zero stored pressure. Allow hot surfaces to cool. Wear hearing, eye and hand PPE.

2. Troubleshooting

Low Discharge Pressure

SAFETY PREREQUISITES

Shut down the compressor. Isolate electrical power and apply lockout/tagout. Close the outlet valve. Vent receiver pressure to zero and verify zero stored pressure. Allow hot surfaces to cool. Wear hearing, eye and hand PPE.

Observed symptoms

The receiver does not reach 6.5 bar, pressure drops during normal demand, the compressor runs continuously, or air leakage can be heard.

Possible causes listed in this manual

Documented causes are an obstructed inlet filter, open drain valve, external air leak, separator restriction, drive problem, or demand above rated capacity.

Inspection guidance

After isolation and full depressurization, inspect the inlet filter, drain valve, hose connections and fittings. Use an approved leak-detection solution. Do not increase the pressure setting above the nameplate rating.

Stop / refusal condition

If the safety prerequisites cannot be confirmed, stop. The manual does not authorize diagnosis or repair without isolation, zero-energy verification and required PPE.

3. Authorized Inspection Procedure

SAFETY PREREQUISITES

Shut down the compressor. Isolate electrical power and apply lockout/tagout. Close the outlet valve. Vent receiver pressure to zero and verify zero stored pressure. Allow hot surfaces to cool. Wear hearing, eye and hand PPE.

- Step 1. Record receiver and discharge pressure.
- Step 2. Stop, isolate and lock out the compressor.
- Step 3. Vent pressure to zero before opening any component.
- Step 4. Inspect inlet filter and drain valve.
- Step 5. Check external connections using approved leak-detection solution.
- Step 6. Inspect cooler airflow and separator condition.
- Step 7. Restore settings only to values listed in this manual.

Reference Parts

AC50-FLT-01	Inlet filter element
AC50-SEP-02	Oil separator element
AC50-VLV-05	Automatic drain valve
AC50-GSK-10	Service gasket kit

All work orders generated from this document require explicit supervisor review and approval before persistence or execution.